FINAL RESEARCH PROJECT - PROCESS REPORT

Project Progress and Research Gap

A stage-gated view of what has been proven, what remains open, and where HPC changes the feasible research scope.

29.7%	503	29.45M
CANONICAL GATE COMPLETION	PLANNED MODEL RUNS	PLANNED TRAINING TIMESTEPS

CURRENT PROJECT BOUNDARY

Canonical completion is 11 of 37 gates (29.7%). The active storage-safe graph has 9 VERIFIED, 1 READY, and 25 PENDING stages. Full screening remains unsubmitted.

TECHNICAL SUMMARY

The project has moved beyond code preparation into a verified HPC execution boundary: observed market data, preregistration, deployment, calibration lineage, preflight, and a five-algorithm smoke run are accepted. The main computational campaign has not started. The immediate research opportunity is to convert this verified platform into a statistically defensible comparison of five reinforcement-learning algorithms under Thai market conditions.

Campaign	large_scale_thai_drl_v2
Evidence cut	17 July 2026
Generated	20 July 2026
Audience	Research team, technical reviewer, and HPC operator

Evidence basis: campaign state, deployment and preflight receipts, job 3550 smoke evidence, and the recorded full local validation run.

SECTION 1

Research question, gap, and evidence boundary

The practical question is whether a learning-based trading policy can produce robust, risk-aware decisions on Thai equities after transaction costs and market-regime changes. The scientific gap is not another single backtest. It is a controlled, repeated comparison that separates algorithm choice, training budget, data context, robustness, and execution stress.

RESEARCH GAP THAT CAN NOW BE FULFILLED

Use the verified storage-safe HPC pipeline to run repeated, preregistered trials for PPO, A2C, DDPG, TD3, and SAC, then test whether any apparent advantage survives nested walk-forward evaluation, robustness checks, feature ablation, and realistic execution stress.

WHY HPC IS MATERIAL

- The full plan contains 503 model runs and 29,450,240 requested training timesteps; a laptop-only workflow would make systematic repetition and validation slow and fragile.
- HPC provides isolated jobs, resource accounting, resumable arrays, and centralized evidence, making failures and results traceable instead of anecdotal.
- More compute does not remove scientific controls. Preregistration, data separation, fixed metrics, deterministic receipts, and leakage checks remain the decision gates.

PROGRESS MEASUREMENT

Canonical continuity — 11/37 verified (29.7%)

Active storage-safe graph — 9/35 verified (25.7%)

WHY TWO PERCENTAGES APPEAR

The canonical 11/37 measure preserves continuity with earlier reports. The active graph has 35 stages because three already verified calibration gates were consolidated into one lineage gate during the storage-safe cutover. Therefore, 9/35 is a structural representation, not a loss of completed work.

SECTION 2

What is complete and what is not

State	Count	Meaning	Current stages
VERIFIED	9	Evidence passed the declared gate.	data_readiness, capacity, preregistration, deployment_bundle, calibration_lineage, screening_config, screening_deployment_bundle, screening_cluster_preflight, screening_smoke
READY	1	Dependencies are verified; execution is allowed only after a fresh operator check.	screening
PENDING	25	Waiting on upstream evidence.	screening_aggregate, budget_config, budget_deployment_bundle, budget_cluster_preflight, budget_sweep, budget_aggregate, nested_config, nested_deployment_bundle, nested_cluster_preflight, nested_screening, plus 15 later stages

VERIFIED TECHNICAL BOUNDARY

- Observed-market feature input is execution-eligible and bound by SHA-256 b877b08ed7a0...291ac6cc.
- The active cutover release contains 248 verified files totaling 7,850,129 bytes.
- Smoke job 3550 completed five algorithms, emitted 5 complete manifests and 5 retention receipts, and imported with status pass.
- The last full local validation recorded 293 passing tests in 36.96 seconds.

NOT YET A SCIENTIFIC RESULT

- The smoke run proves executability and storage behavior; it is too small to support algorithm-ranking or investment-performance claims.
- The 150-trial screening array and its aggregate have not been submitted.
- Nested evaluation, robustness, feature ablation, execution stress, and final statistical synthesis remain pending.

SECTION 3

Interpretation, limitations, and next questions

KEY FINDINGS

- The project has a reproducible execution platform, not a completed empirical answer.
- The storage-safe design cut observed smoke output by 49.94% while preserving final models and evidence.
- The next defensible milestone is full screening plus aggregation, using a new quota snapshot and a deliberately small submission window.

LIMITATIONS AND ROBUSTNESS BOUNDARY

- The current observed dataset is a restricted private research pilot; findings must be described within that evidence class.
- Runtime projections are extrapolations from one short smoke trial per algorithm and include a 1.5 safety margin; they are planning evidence, not guarantees.
- The post-smoke group soft-quota margin was only about 0.309 GiB. That point-in-time pass expired after 15 minutes and cannot authorize a later full run.
- Macro and fundamental feature claims remain outside scope until observed inputs are checksum- and readiness-certified, or the preregistration is amended before test access.

QUESTIONS THE NEXT PHASE SHOULD ANSWER

- Which algorithms are consistently competitive after costs across time splits, rather than on one favorable period?
- How sensitive are conclusions to training budget and hyperparameter selection?
- Do advantages survive nested walk-forward testing and multiple-comparison control?
- Which data contexts and execution assumptions materially change the result?

DECISION

Proceed to a fresh capacity audit and a bounded screening window. Do not describe the project as complete, and do not submit full screening from the expired 17 July quota decision.

PRIMARY EVIDENCE

Artifact or identifier	Verified value
Active campaign state	`campaigns/large_scale_thai_drl_v2/state.json`
Storage-safe smoke summary	`results/screening_storage_safe_smoke_3550/screening_smoke_summary.json`
Smoke import validation	`reports/screening_storage_safe_smoke_3550_import_validation.json`
Cutover preflight	`reports/bistka_screening_storage_safe_cutover_v2_preflight.json`